METHOD 23 / A COMPLETE BEGINNER GUIDE

DCCA detrended cross-correlation

Question this method answers: After building cumulative return profiles and removing local trends, how strongly do two FX series co-move at window scale s?

ρ_DCCA(s) differs from raw-return Pearson r — detrending inside each block targets long-memory co-fluctuations.

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Teaching returns only — no snapshot DCCA scaling plots.

Prepared for: 23 - DCCA Detrended Cross-Correlation. README status: not started.

02 / THE BASIC IDEA

Long-memory co-movement

Pearson r on daily returns treats each day independently in the formula. DCCA integrates returns into profiles, removes polynomial trends in sliding windows, then correlates residual profiles — emphasizing slow co-movement across scales s.

Different object

Pearson on returns	One-step linear linkage.
ρ_DCCA(s)	Detrended profile correlation at window length s.
Scale sweep	Compare s = 20 versus s = 100 observations (formula sheet).

03 / RETURNS INTEGRATED

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

1.1000 → 1.1110 is about +1% simple return. DCCA starts from demeaned returns x_t – x■, then cumulates. Tiny daily percents stack into profile shapes over weeks.

Use the same return type for both legs; mixing simple and log breaks comparability.

04 / CUMULATIVE SUM

Build the profile

```
X_k = sum_{t=1..k} ( x_t - x_bar )
Y_k = sum_{t=1..k} ( y_t - y_bar )
```

Profiles look like rough random walks built from returns. Local linear (or polynomial) trends inside each window of size s are fitted and subtracted to obtain X■ and ■ segments.

Section 23 in forex-formulas-explained.pdf uses the same mean-of-products normalization shown on the next pages.

05 / ZEBENDE (2011); KWAPIE■ ET AL. (2015)

Research notes

Zebende (2011): Detrended cross-correlation analysis of two nonstationary time series. Physica A, 390, 2480–2490 — introduces ρ_DCCA scaling across window sizes.

Kwapie■, O■wi■cimka, Dro■d■ (2015): Detrended fluctuation analysis made flexible — extends DFA/DCCA family including q-dependent variants (method 24 in this repo).

FX applications must specify trend order, overlap, and averaging — defaults from physics papers may not suit daily FX without review.

06 / SMALL MATH DICTIONARY

Symbol dictionary

Symbol	Meaning
x_t, y_t	Demeaned daily returns.
X_k, Y_k	Cumulative profiles.
s	Detrending window length (observations).
X■, ■	Profile residuals after local detrend.
ρ_DCCA(s)	Detrended cross-correlation at scale s.
Overlap	How windows slide (non-overlap versus 50%).

07 / LOCAL LINEAR FIT

Detrend inside each window

Split the profile into consecutive (or overlapping) segments of length s. In each segment, fit a line (or low-order polynomial) to X and separately to Y. Subtract the fit to get residuals for that segment.

Concatenate residual segments (or average overlapping estimates) before the correlation step. Document overlap — ρ_DCCA changes with it.

08 / NORMALIZE COVARIATION

ρ_DCCA formula

```
rho_DCCA(s) = mean(X_tilde * Y_tilde) /
sqrt( mean(X_tilde^2) * mean(Y_tilde^2) )
```

Means are taken over all residual points contributed at scale s. The score behaves like a correlation on detrended profile fluctuations — not on raw returns.

```
Teaching window s = 5: rho_DCCA ≈ 0.9989
Raw Pearson on returns ≈ 0.9992
```

The two numbers need not match — that gap is expected when slow profile co-movement differs from one-day linear linkage.

Compare scales

Repeat ρ_DCCA(s) for several s values and plot log(s) versus ρ or related fluctuation functions. FX crisis periods may show stronger detrended linkage at medium s than calm periods.

Method 24 (multifractal DCCA) generalizes with exponent q — stay with q = 2 / standard DCCA here unless you read method 24.

10 / REPO USAGE

Daily majors workflow

1. Long aligned return panels

DCCA needs enough length for multiple windows — short snapshots give noisy scales.

2. Freeze detrend degree and overlap

Log choices in experiments/v001/ NOTES.

3. Report ρ_DCCA(s) curves

Do not reduce to a single s tuned after seeing pair rankings.

Versus Pearson and rolling r

Rolling Pearson (method 05) tracks time-varying linear correlation on returns. DCCA emphasizes detrended profile structure across scales — closer to long-memory diagnostics.

Neither replaces tail counts (19) or wavelet maps (22).

If ρ_DCCA(s) rises while rolling r falls, slow profile co-movement may persist after short-horizon decorrelation — but verify with stable window counts and overlapping rules before interpreting.

Document whether profiles were built from log or simple returns; integration makes small differences in x■ matter over long panels.

Software sketch

```
# profile = cumsum(x - mean(x))
# for each window: polyfit trend, subtract
# correlate flattened residuals
```

Libraries: nolds, MFDFA implementations, or custom code — verify mean normalization matches Zebende (2011) before comparing to literature.

Keep DCCA code inside method experiments; shared/mt5_data stays I/O only.

Limits

Nonstationary FX regimes break fixed-scale interpretations. Polynomial detrend order can absorb real co-movement if set too high.

ρ_DCCA(s) is not a trading signal, not TE, and not a copula tail measure.

Very small samples yield too few windows at large s — drop unreliable scales instead of extrapolating.

Practice

Q: Is ρ_DCCA(s) equal to Pearson r? **A:** Not in general — different objects (see teaching numbers on page 8).

Q: Can s = 5 on 20 days support s = 100 analysis? **A:** No — you need enough profile length.

Check: teaching ρ_DCCA(5) ≈ 0.9989.

Q: Do profiles use price levels? **A:** No — integrate demeaned returns.

Sources

[1] Zebende, G. F. (2011). Detrended cross-correlation analysis of two nonstationary time series. Physica A, 390, 2480–2490.
ρ_DCCA definition.
Open source

[2] Kwapie■, J., O■wi■cimka, P., & Dro■d■, S. (2015). Detrended fluctuation analysis made flexible. Physica A, 427, 1–8.
Flexible DFA/DCCA framework.
Open source

[3] forex-formulas-explained.pdf — section 23 (this repository).
Project formula sheet.
Open source

Rebuild: python build_guide.py. End of DCCA detrended cross-correlation beginner guide.